



University
of Exeter

COMM113

Deep Learning

Lecturer: Gaojie (Jay) Jin

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What this course is about

- Core concepts of Deep Learning (DL),
- How to improve generalisation and robustness of DL models
- Strengths and weaknesses of different methods
- All at a conceptual level... mostly

Assessment:

- Two assessments:
 - Coursework 1, worth 30% of your mark due 17 November 2025
(MLP on FashionMNIST using Pytorch (30%) and Numpy-only (70%))
 - Written Exam, worth 70% of your mark
(Conceptual:60%, Computation:40%, e.g., backpropagation)
 - More detail to come!

Lectures:

- Lectures:
 - Fridays 8:35 AM - 9:25 AM
203 /Harrison
 - Fridays 4:35 PM - 5:25 PM
LT4.2 /Queens
 - There might be changes. Check your timetable.
- Recorded and posted on ele.exeter.ac.uk

Practical workshops:

- Optional, but **very** useful
- You can work in groups and get help from the teaching assistants
 - **Monday 8:35 AM-9:25 AM**, 207 /Harrison

The normal workshop will start in week 2.

- TA: Jay g.jin@Exeter.ac.uk

Resources

Textbooks (Foundational & Practical)

- Deep Learning (Goodfellow, Bengio, Courville, 2016) – classic reference for theory & foundations. <https://www.deeplearningbook.org/>
- Dive into Deep Learning (Zhang et al., 2023, open-source) – interactive book with code in PyTorch/MXNet. <https://d2l.ai>
- Neural Networks and Deep Learning (Michael Nielsen, free online book) – beginner-friendly introduction. <http://neuralnetworksanddeeplearning.com>

Framework-Specific (PyTorch Focused)

- PyTorch Official Tutorials – hands-on beginner to advanced guides. <https://pytorch.org/tutorials>
- Deep Learning with PyTorch (Eli Stevens, Luca Antiga, Thomas Viehmann, 2020) – practical book with PyTorch focus.
- PyTorch Lightning Docs – structured PyTorch training framework. <https://lightning.ai/docs>

Online Courses

- CS231n: Convolutional Neural Networks for Visual Recognition (Stanford) – highly recommended for computer vision. <http://cs231n.stanford.edu>
- Deep Learning Specialization (Andrew Ng, Coursera) – covers fundamentals, practical coding.
- Fast.ai Practical Deep Learning for Coders – very hands-on, PyTorch-based. <https://course.fast.ai>

Week 1 – Introduction to Deep Learning

What is Deep Learning?

- Good old Neural Networks, with more layers/modules
- Non-linear, hierarchical, abstract representations of data
- Flexible models with any input/output type and size
- Differentiable Functional Programming

If you had to explain deep learning to your parents in one sentence, how would you do it?

What is Deep Learning?

Deep learning is a class of machine learning algorithms that:

- use a cascade of multiple layers of nonlinear processing units for feature extraction and transformation. Each successive layer uses the output from the previous layer as input.
- learn in supervised (e.g., classification) and/or unsupervised (e.g., pattern analysis) manners.
- learn multiple levels of representations that correspond to different levels of abstraction.

Why Deep Learning Now

Limitations of traditional machine learning algorithms

- not good at handling high dimensional data.
- difficult to do feature extraction and object recognition.

Advantages of deep learning

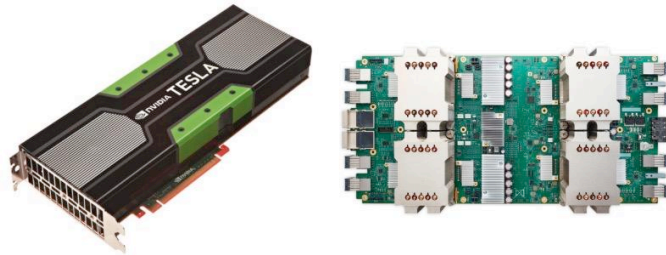
- DL is computationally expensive, but it is capable of handling high dimensional data.
- feature extraction is done automatically.

Why Deep Learning Now

- Better algorithms & understanding
- Computing power (GPUs, TPUs, ...)
- Data with labels
- Open source tools and models

Why Deep Learning Now

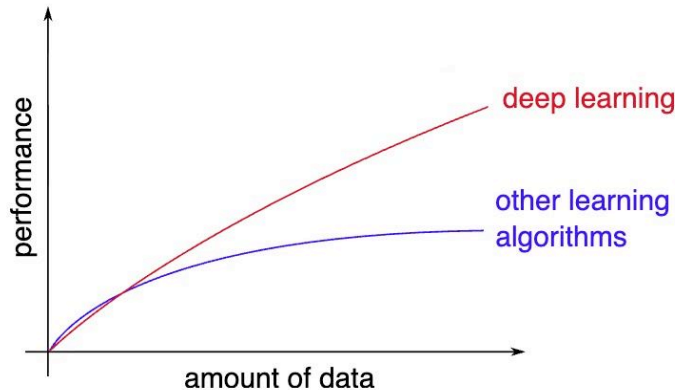
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GPU and TPU

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PYTORCH



Microsoft
CNTK



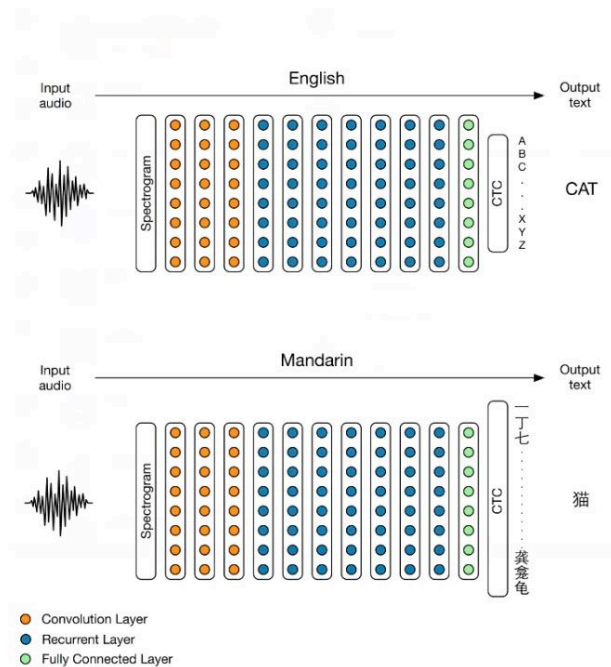
dmlc
mxnet

gensim

spaCy

theano

DL Today: Speech-to-Text



[Baidu 2014]

How many of you have used voice typing on your phone? Did it work well? What went wrong?

DL Today: Vision



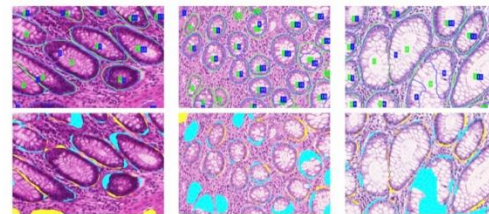
[Krizhevsky 2012]



[Ciresan et al. 2013]



[Stanford 2017]

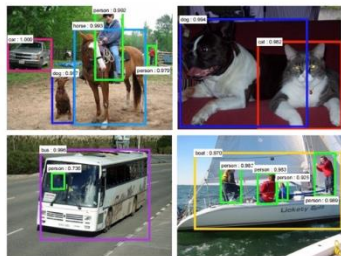


(d) benign

(e) benign

(f) malignant

[Nvidia Dev Blog 2017]



[Faster R-CNN - Ren 2015]

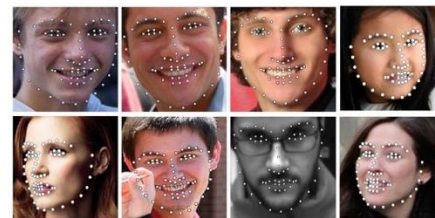


[NVIDIA dev blog]



Figure 1. Illumination and Pose invariance.

[FaceNet - Google 2015]



[Facial landmark detection CUHK 2014]

What's the most surprising use of computer vision you've heard of?

DL Today: NLP

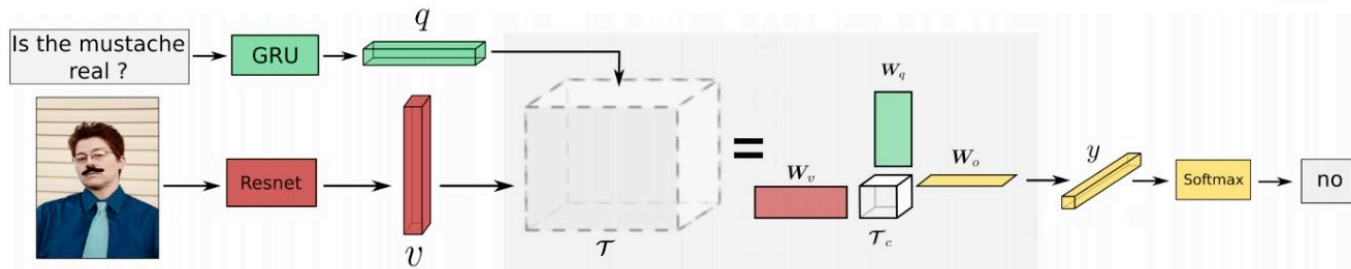


[Google Inbox Smart Reply]



[Amazon Echo / Alexa]

DL Today: Vision+NLP

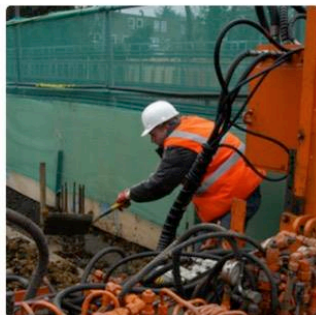


[VQA - Mutan 2017]

Imagine taking a photo and asking your phone: 'What's happening here?' How accurate do you think it would be today?



"man in black shirt is playing guitar."



"construction worker in orange safety vest is working on road."



"two young girls are playing with lego toy."



"boy is doing backflip on wakeboard."

[Karpathy 2015]

[Real-time webcam demo](#)

DL Today: Generative Models

When training begins, the generator produces obviously fake data, and the discriminator quickly learns to tell that it's fake:

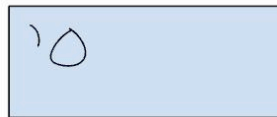
Generated Data

Discriminator

Real Data



Sampled celebrities [Nvidia 2017]



FAKE

REAL



As training progresses, the generator gets closer to producing output that can fool the discriminator:



FAKE

REAL



Finally, if generator training goes well, the discriminator gets worse at telling the difference between real and fake. It starts to classify fake data as real, and its accuracy decreases.



REAL

REAL



StackGAN v2 [Zhang 2017]

Have you ever seen an AI-generated picture and thought it was real?

Demo: [Drag your GAN](#)

DL Today: LLMs/VLMS

Open-AI GPT-3, or DALL-E: <https://openai.com/blog/dall-e/>

TEXT PROMPT

an armchair in the shape of an avocado [...]

AI-GENERATED IMAGES



[View more or edit prompt ↕](#)

TEXT PROMPT

a store front that has the word 'openai' written on it [...]

AI-GENERATED IMAGES

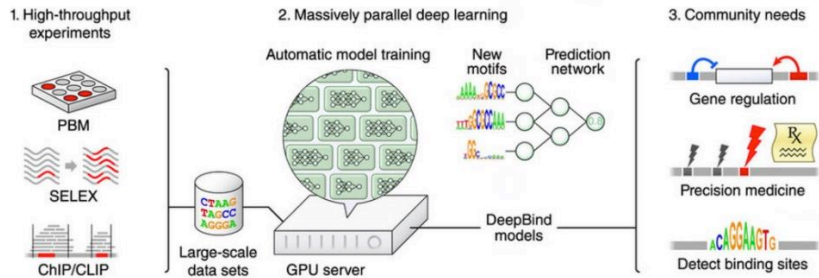


[View more or edit prompt ↕](#)

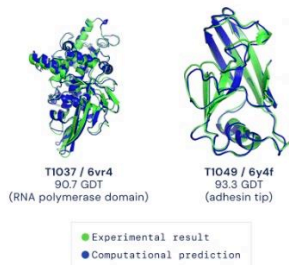
[AI Generated Movies](#)

What risks do you see if we start relying too much on these systems for decision-making or learning?

DL Today: Science



[Deep Genomics 2017]



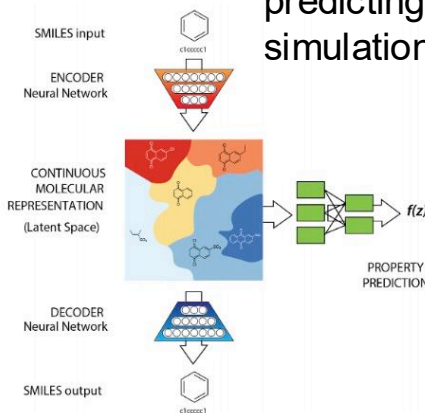
[AlphaFold by DeepMind](#)

Why do you think AI is so useful in science — is it the speed, the scale, or something else?



[NVIDIA ALCHEMI](#): AI for Chemistry and Materials Science

predicting molecular properties faster than traditional simulations

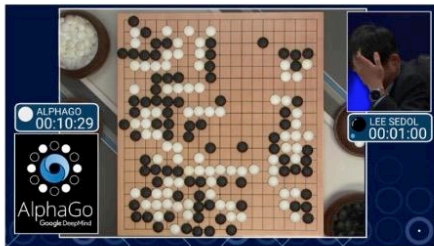


[Gómez-Bombarelli 2016]

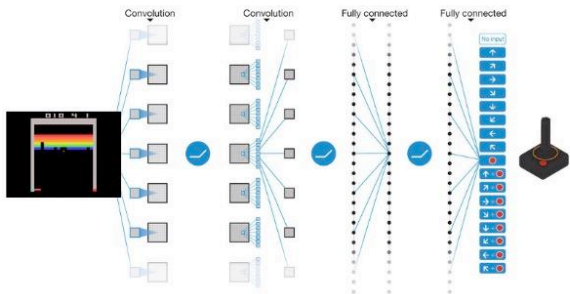
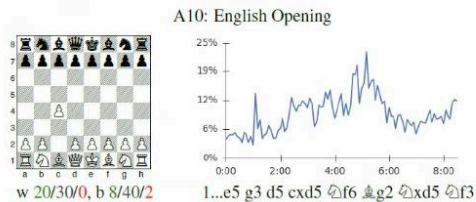


[Tompson 2016]

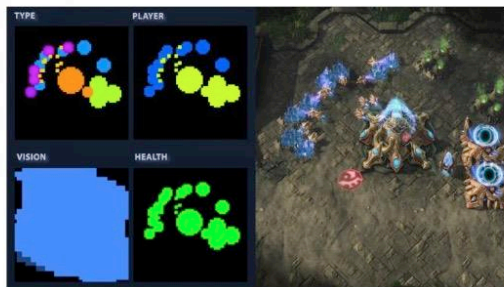
DL Today: Games



[Deepmind AlphaGo / Zero 2017]



[Atari Games - DeepMind 2016]

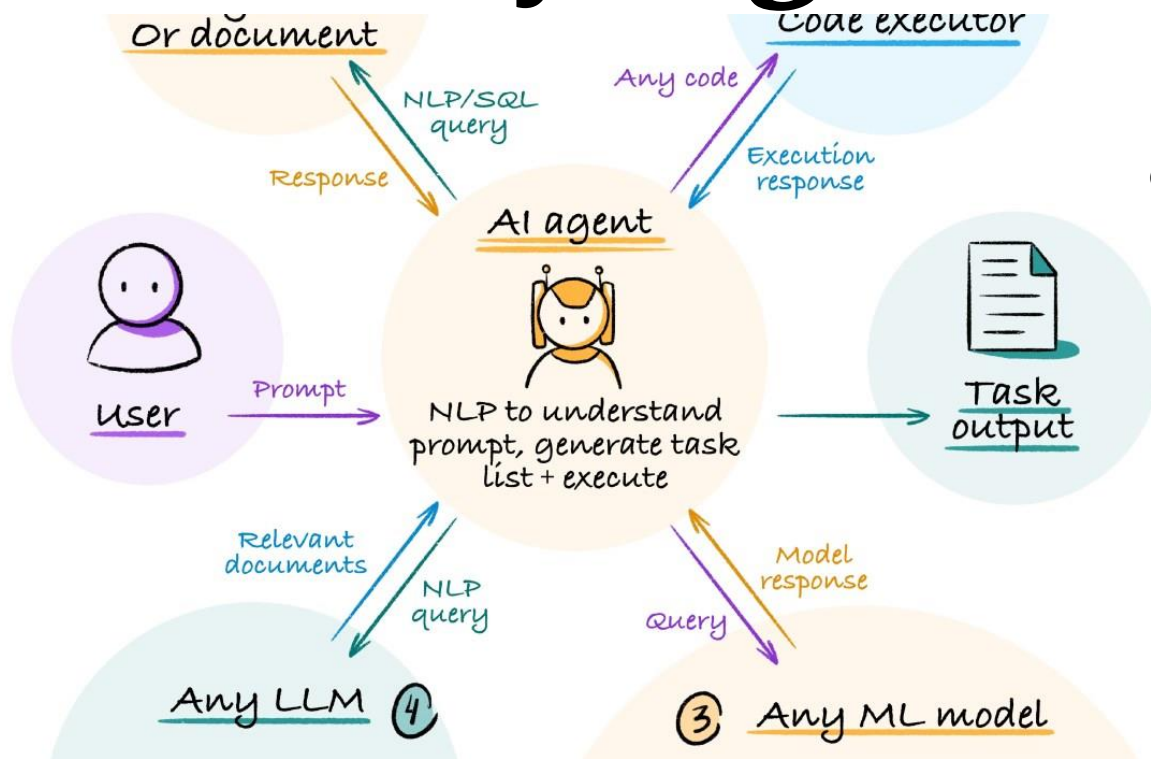


[Starcraft 2 for AI research]



Has anyone here followed AlphaGo or AlphaStar? Why do you think beating humans at these games was considered such a big deal?

DL Today: Agentic AI



We're moving toward agentic AI

don't just respond but *act* in environments

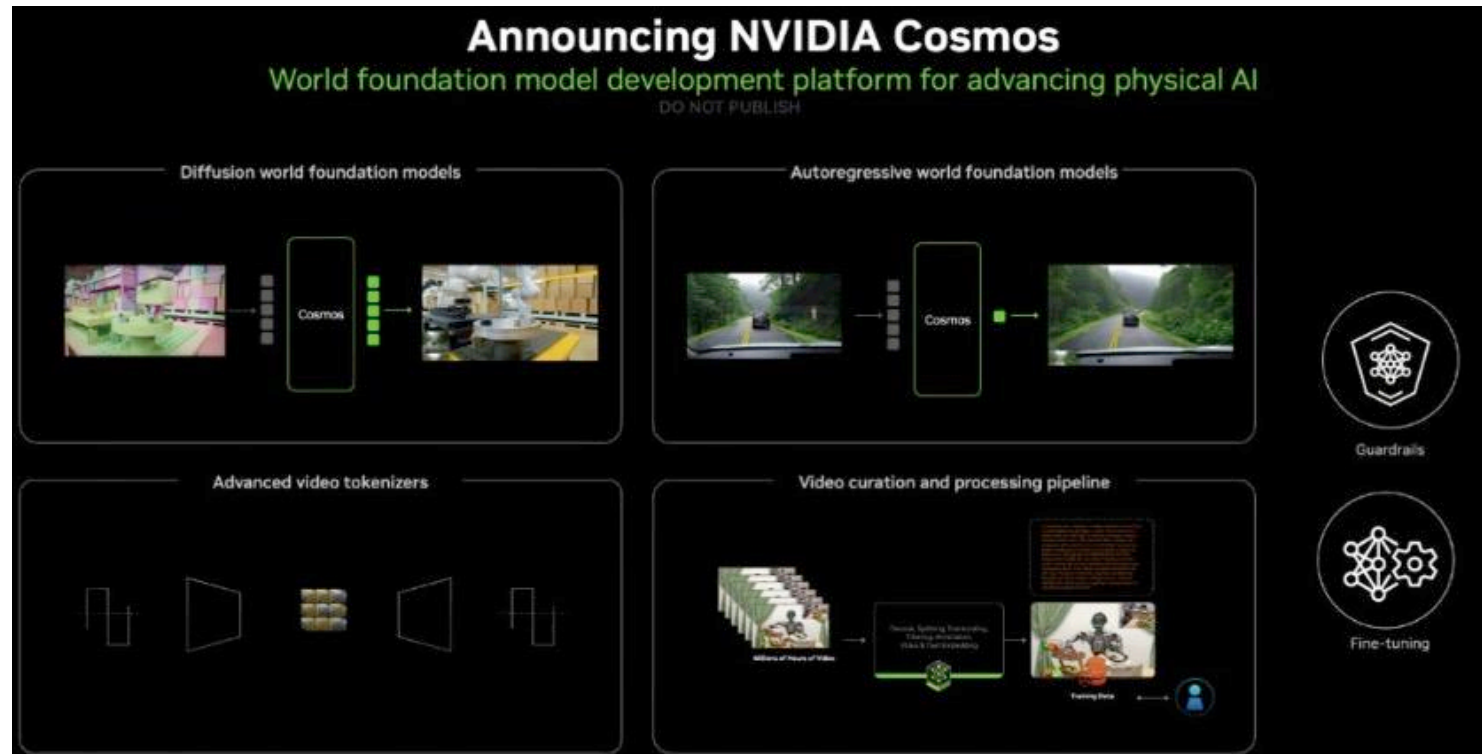
combine reasoning, memory, and interaction — much closer to how humans operate.

If you had an AI agent as a study buddy, what tasks would you want it to do for you?

Build an AI agent as teaching assistant

World Models

World foundation models are neural networks that simulate real-world environments as videos and predict accurate outcomes based on text, image, or video input.



For example, NVIDIA Cosmos builds world foundation models to simulate physical systems — almost like a digital twin of reality.

[NVIDIA Cosmos](#)

If we can build accurate world models, what industries could change the most? Healthcare? Climate? Robotics?